

Test 3

March 12, 2024

Problem 1

Factor out the greatest common factor from the expression $28b + 14b^2 + 35b^3 + 7b^5$.

Problem 2

Factor the expression $3n^3 - 2n^2 - 9n + 6$.

Problem 3

Factor the expression $x^2 - 7x - 18$.

Problem 4

Factor the following expressions:

a) $a^2 - b^2$

b) $a^4 - b^4$

c) $x^4 - 16$

d) $4x^2 - 36$

Problem 5

Solve the expression $x^2 - 8x - 3 = -3$.

Problem 6

Simplify the expression (hint: the numerator is NOT 0).

$$\frac{x+1}{x^2+8x+7}$$

Problem 7

Simplify the expression:

$$\frac{6x(x+4)}{x-3} \cdot \frac{(x-3)(x-6)}{6x(x-6)}$$

Problem 8

Find the missing expression:

$$\frac{x-4}{x+2} \cdot \frac{?}{x^2+5x+6}$$

Problem 9

Add and simplify the rational expressions:

$$\frac{x+5}{8} + \frac{x-3}{12}$$

Problem 10

Solve:

$$\frac{\frac{a^2-b^2}{4a^2b}}{\frac{a+b}{16ab^2}}$$

Problem 11

Solve for x :

$$\frac{8}{5} = \frac{3}{x-8}$$

Problem 12

Solve for x :

$$\frac{1}{x+2} + \frac{1}{x-2} = \frac{3x+8}{x^2-4}.$$

Problem 13

Convert 435,000 m^2 to square kilometers.

Problem 14

Simplify: $8\sqrt{98 \cdot m \cdot n}$.

Problem 15

Simplify: $8\sqrt[3]{64 \cdot u^5 \cdot v^3}$.

Problem 16

Simplify: $3\sqrt{2} + 2\sqrt{8} - 3\sqrt{18}$.

Problem 17

Simplify:

$$\frac{4\sqrt{3}}{\sqrt{15}}$$

Problem 18

Simplify:

$$\frac{3}{4 - 3\sqrt{3}}$$

Problem 19

Simplify. Your answer should contain only positive exponents.

$$\frac{x^2 y^{\frac{2}{3}} \cdot 2x^{\frac{1}{2}} y^{\frac{5}{6}}}{x^{\frac{7}{2}} y^0}$$

Extra Credit

Simplify:

$$\frac{8i}{6 - 7i}$$